COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 Data Structures and Algorithms	**SECTION:**	CPE21S1
DATE PERFORMED:	7/28/26	**DATE SUBMITTED:**	7/28/26
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ACTIVITY NO. ___ [ACTIVITY NAME]

1. PROCEDURE OUTPUT

ILO A

CODE

```cpp
#include <iostream>
#include <stack> //calling stack from stl

int main(){
    //create an object for the stack
    std::stack<int> newStack;

    std::cout<<"Stack Empty? : "<<newStack.empty()<<std::endl;

    //push to the stack
    newStack.push(3);
    newStack.push(8);
    newStack.push(15);

    std::cout<<"Stack Empty? : "<<newStack.empty()<<std::endl;
    std::cout<<"The size of the stack is : "<<newStack.size()<<std::endl;
    std::cout<<"The top of the element is: "<<newStack.top()<<std::endl;

    std::cout<<"Removing top element from the stack : "<<newStack.top()<<std::endl;
    newStack.pop();

    std::cout<<"Removing top element from the stack : "<<newStack.top()<<std::endl;
    newStack.pop();

    std::cout<<"Removing top element from the stack : "<<newStack.top()<<std::endl;
    newStack.pop();

    return 0;
}
```

OUTPUT

```
Stack Empty? : 1
Stack Empty? : 0
The size of the stack is : 3
The top of the element is: 15
Removing top element from the stack : 15
Removing top element from the stack : 8
Removing top element from the stack : 3
```

ILO B

```cpp
#include <iostream>

//static array, what do we need?
const size_t maxCap = 100;
int stack[maxCap];
int top = -1, i, newData, choice;

void push();
void pop();
void Top();
void display();
bool isEmpty();
bool isFull();


int main(){

    std::cout<<"\nEnter the number of max elements for the new stack: ";
    std::cin>>i;

    while(true){
        std::cout<<"\n--Stack operations--\n"<<std::endl;
        std::cout<<"1. Push, 2. Pop, 3. Top, 4. IsEmpty 5. Display Elements\n\n Choose: ";
        std::cin>>choice;
        switch (choice){

            case 1: push();
            break;
            case 2: pop();
            break;
            case 3: Top();
            break;
            case 4: std::cout<< isEmpty()<<std::endl;
            break;
            case 5: display();
            break;
            default : std::cout<<"\n========Invalid choice========"<<std::endl;
            break;
            }

    }
    return 0;
```

```
}
bool isEmpty(){
    //create a condition to check wheter the top element is equal to -1
    if(top == -1) return true;
    return false;
}
bool isFull(){
    //check if we reach the maximunm capacity of the array
    if(top >= (int)maxCap - 1) return true;
    return false;
}
void push(){
    //check if full - > if yes return an error
    if(top == i - 1){
    std::cout << "\n ==============STACK OVERFLOW==============" << std::endl;
    return;
    }

    std::cout<<"\n New value to add into the stack: ";
    std::cin>>newData;
    stack[++top]=newData;
}
void Top(){

    std::cout<<"\nThe element on the top of the stack is: "<<stack[top]<<std::endl;
}
```

```
void pop(){
    //check if the stack is empty
    if(isEmpty()){
        std::cout<<"\n==============STACK UNDERFLOW=============="<<std::endl;
        return;
    }
    std::cout << "\n Popping: "<< stack[top--]<<std::endl;
}
void display(){
    if(isEmpty()){
        std::cout<<"\n========STACK IS EMPTY========"<<std::endl;
        return;
    }
    std::cout<<"\nSTACK ELEMENTS(bottom to top) : ";
    for(int j = 0; j<=top; j++){
        std::cout<<stack[j]<<" ";
    }
    std::cout<<std::endl;
}
```

CODE OUTPUT

```
Enter the number of max elements for the new stack: 3

--Stack operations--

1. Push, 2. Pop, 3. Top, 4. IsEmpty 5. Display Elements

 Choose: 5

========STACK IS EMPTY========
```

```
--Stack operations--

1. Push, 2. Pop, 3. Top, 4. IsEmpty 5. Display Elements

 Choose: 4
1
```

```
1. Push, 2. Pop, 3. Top, 4. IsEmpty 5. Display Elements

 Choose: 1

 New value to add into the stack: 5

--Stack operations--

1. Push, 2. Pop, 3. Top, 4. IsEmpty 5. Display Elements

 Choose: 1

 New value to add into the stack: 6

--Stack operations--

1. Push, 2. Pop, 3. Top, 4. IsEmpty 5. Display Elements

 Choose: 1

 New value to add into the stack: 7

--Stack operations--

1. Push, 2. Pop, 3. Top, 4. IsEmpty 5. Display Elements

 Choose: 1

 ==============STACK OVERFLOW==============
```

```
--Stack operations--

1. Push, 2. Pop, 3. Top, 4. IsEmpty 5. Display Elements

 Choose: 2

 Popping: 7

--Stack operations--

1. Push, 2. Pop, 3. Top, 4. IsEmpty 5. Display Elements

 Choose: 1

 New value to add into the stack: 8

--Stack operations--

1. Push, 2. Pop, 3. Top, 4. IsEmpty 5. Display Elements

 Choose: 1

 ==============STACK OVERFLOW===============

--Stack operations--

1. Push, 2. Pop, 3. Top, 4. IsEmpty 5. Display Elements

 Choose: 5

STACK ELEMENTS(bottom to top) : 5 6 8

--Stack operations--

1. Push, 2. Pop, 3. Top, 4. IsEmpty 5. Display Elements

 Choose:
```

ILO B2

MAIN CPP

```cpp
#include <iostream>
#include "stacklist.h"

int main(){

    std::cout<<"--------PUSHING 'c' 'p' 'e' ONTO THE STACK --------\n";

    //push
    push('c');
    push('p');
    push('e');

    //display the current top
    Top<char>();// this should be 'e'

    std::cout<<"\n -----Popping all the elements-----\n";

    while(!isEmpty<char>()){
        std::cout<<"popped: "<< pop<char>()<<std::endl;
    }
    std::cout<<"\n -----Testing the Underflow-----\n";
    pop<char>();


    return 0;
}
```

HEADER FILE

```cpp
#ifndef STACK_LIST_H
#define STACK_LIST_H


//class node that contains data and the next
template <typename T>
class Node{
    public:
    T data;         //value that we need to store
    Node* next;     //pointer to the next node

};


//global pointer for the top of the stack
template <typename T>
Node<T> *head = nullptr;

//operations for the stack

//push
template <typename T>
void push(T newData){
    //create a new node and assign the newData
    Node<T>* newNode = new Node<T>;
    newNode->data= newData;

    //new node points to the current top, then becomes the new top
    newNode->next = head<T>;
    head<T> = newNode;
}
```

```
34  T pop(){
35      //checks if the stack is empty
36      if(head<T> == nullptr){
37          std::cout<<"Stack Underflow."<<std::endl;
38          return T();
39      }
40      //save the current top node
41      Node<T>* temp = head<T>;
42      //store its data
43      T tempVal = temp -> data;
44
45      //move the head to the next node
46      head<T> = head<T>->next;
47
48      //free the removed node
49      delete temp;
50
51      //return the removed value
52      return tempVal;
53  }
54  //isEmpty
55  template <typename T>
56  bool isEmpty(){
57      return head<T> == nullptr;
58  }
59  //top
60  template <typename T>
61  void Top(){
62      // error catching
63      if(head<T> == nullptr){
64          std::cout<<"Stack Underflow"<<std::endl;
65          return;
66      }
```

```
67      //display the top Node
68      std::cout<<"the top tof the stack is: '"<<head<T>->data<<"'"<<std::endl;
69  }
70
71  template<typename T>
72  void displayAll(){
73      if(head<T> == nullptr){
74          std::cout<<"stack underflow\n";
75          return;
76      }
77      Node<T>* temp = head<T>;
78      while(temp != nullptr){
79          std::cout<<temp->data << " ";
80          temp = temp->next;
81      }
82      std::cout<<"\n";
83  }
84
85
86
87
88  #endif
```

CODE OUTPUT

```
--------PUSHING 'c' 'p' 'e' ONTO THE STACK --------
the top tof the stack is: 'e'

 -----Popping all the elements-----
popped: e
popped: p
popped: c

 -----Testing the Underflow-----
Stack Underflow.
```

CONCLUSION:

In this Seatwork i learned how to use and implement array based stack list and its functions and also the stack operations like the transformers which are push and pop and then observers which are top, isFull, and IsEmpty, after learning about it i got to apply all of those in my code.The seatwork was challenging because it is a new topic but i got to learn in as i got to use and apply it to the code. Overall this seatwork was a tough challenge but it also helped me learn so many things

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

ILO C

MAIN CPP

HEADER:

```cpp
#include <iostream>
#include <string>

struct Node{
    char data;
    Node* next;
};

// Top starts as nullptr because the list is empty
Node* top = nullptr;

bool isEmpty(){
    if(top == nullptr) return true;
    return false;
}

void push(char c){
    Node* newNode = new Node();
    newNode->data = c;
    newNode->next = top;
    top = newNode;
}

char pop(){
    if(isEmpty()){
        return '\0';
    }
    Node* temp = top;
    char poppedData = temp->data;
    top = top->next;
    delete temp;
    return poppedData;
}

```

INT MAIN

```cpp
int main(){
    std::string expr;

    std::cout << "Enter expression: ";
    std::cin >> expr;

    bool isValid = true;

    for(int j = 0; j < expr.length(); j++){
        char ch = expr[j];

        if(ch == '(' || ch == '{' || ch == '['){
            push(ch);
        }
        else if(ch == ')' || ch == '}' || ch == ']'){
            if(isEmpty()){
                isValid = false;
                break;
            }

            char popped = pop();

            if((ch == ')' && popped != '(') ||
               (ch == '}' && popped != '{') ||
               (ch == ']' && popped != '[')){
                isValid = false;
                break;
            }
        }
    }

    if(!isEmpty()){
        isValid = false;
    }

    if(isValid){
        std::cout << "Result: Expression is VALID." << std::endl;
    }else {
        std::cout << "Result: Expression is INVALID (Unbalanced Symbols)." << std::endl;
    }

    return 0;
}
```

CODE OUTPUT

```
C:\MinGW\bin>C:\Users\vince\OneDrive\CPPGARCIA\ILOC\main.exe
Enter expression: (A+B)+(C-D)
Result: Expression is VALID.

C:\MinGW\bin>C:\Users\vince\OneDrive\CPPGARCIA\ILOC\main.exe
Enter expression: ((A+B)+(C-D)
Result: Expression is INVALID (Unbalanced Symbols).

C:\MinGW\bin>C:\Users\vince\OneDrive\CPPGARCIA\ILOC\main.exe
Enter expression: ((A+B)+[C-D])
Result: Expression is VALID.

C:\MinGW\bin>C:\Users\vince\OneDrive\CPPGARCIA\ILOC\main.exe
Enter expression: ((A+B]+[C-D]}
Result: Expression is INVALID (Unbalanced Symbols).
```

Expression	Valid? (Y/N)	Output (Console Screenshot)	Analysis
(A+B)+(C-D)	Y	Enter expression: (A+B)+(C-D) Result: Expression is VALID.	The output is valid because all the open parenthesis have close parenthesis
((A+B)+(C-D)	N	Enter expression: ((A+B)+(C-D) Result: Expression is INVALID (Unbalanced Symbols).	The output is invalid because the parenthesis on the first does not have a closed parenthesis
((A+B)+[C-D])	Y	Enter expression: ((A+B)+[C-D]) Result: Expression is VALID.	Same with the first output but this is also valid because the open bracket has a closed bracket
((A+B]+[C-D]}	N	Enter expression: ((A+B]+[C-D]} Result: Expression is INVALID (Unbalanced Symbols).	The curly bracket doesnt have a match at the end which makes it invalid

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

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